

JUNIOR DATA ANALYST PILOT PROJECT SUBMISSION

Project 4: BigQuery + SQL + Looker Studio

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Company	Cellutech FZCO
Dataset	10,000 iPhone Transaction Records 93 SKUs
BigQuery Dataset	cellutech_iphone_sales
Tools Used	Google BigQuery, SQL (Standard), Looker Studio
Date	September 2026

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Submission Checklist

The following table maps each assignment deliverable to the corresponding submission artifact.

Deliverable	Required Submission	Artifact / Reference	Status
BigQuery	Dataset/table/view names or screenshots proving the BigQuery setup	See Section 2 below (screenshots + object names)	Done
SQL	SQL script(s) used for cleaning, formatting, transformations and analysis	pilot_project_sql_scripts.sql SQL_Scripts_Deliverable_Ismail_Dad_Khan.docx	Done
Dashboard	Shareable Looker Studio dashboard connected to BigQuery	Looker Studio link (see Section 4)	Done
Validation	Brief reconciliation showing source vs reporting-layer row counts/totals	See Section 5 below (reconciliation table)	Done
Findings	At least 3 business findings supported by the dashboard/SQL output	See Section 6 below (6 findings with data points)	Done

Deliverable 1: BigQuery Setup

This section documents the BigQuery environment, dataset, staging tables, and reporting view created for this project.

1.1 Data Flow Pipeline

Source Excel Files → BigQuery Staging Tables → SQL Transformation (CTE + JOIN + CASE) → Reporting View → Looker Studio Dashboard

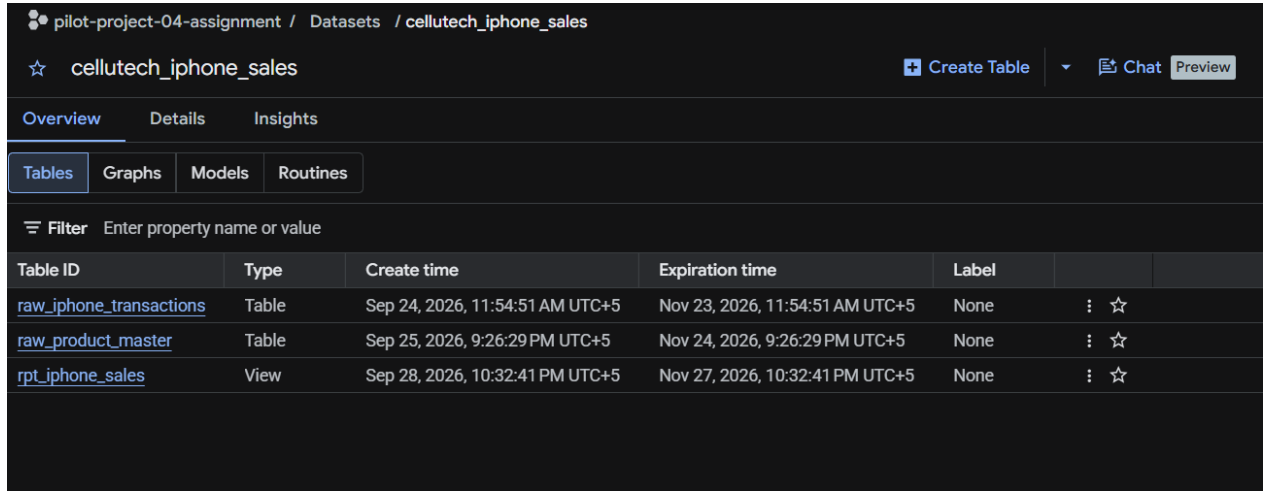
1.2 BigQuery Objects Created

Object Name	Type	Row Count	Description
cellutech_iphone_sales	Dataset	—	BigQuery dataset containing all project objects
raw_iphone_transactions	Staging Table	10,920*	Raw upload from Excel source file
raw_product_master	Reference Table	93	Product dimension table (Model, Storage, Color)
rpt_iphone_sales	Reporting View	10,000	Cleaned, enriched, analysis-ready reporting layer

**10,920 includes 920 trailing blank rows from Excel formatting. These are filtered to 10,000 valid records in the reporting view.*

1.3 BigQuery Console Screenshots

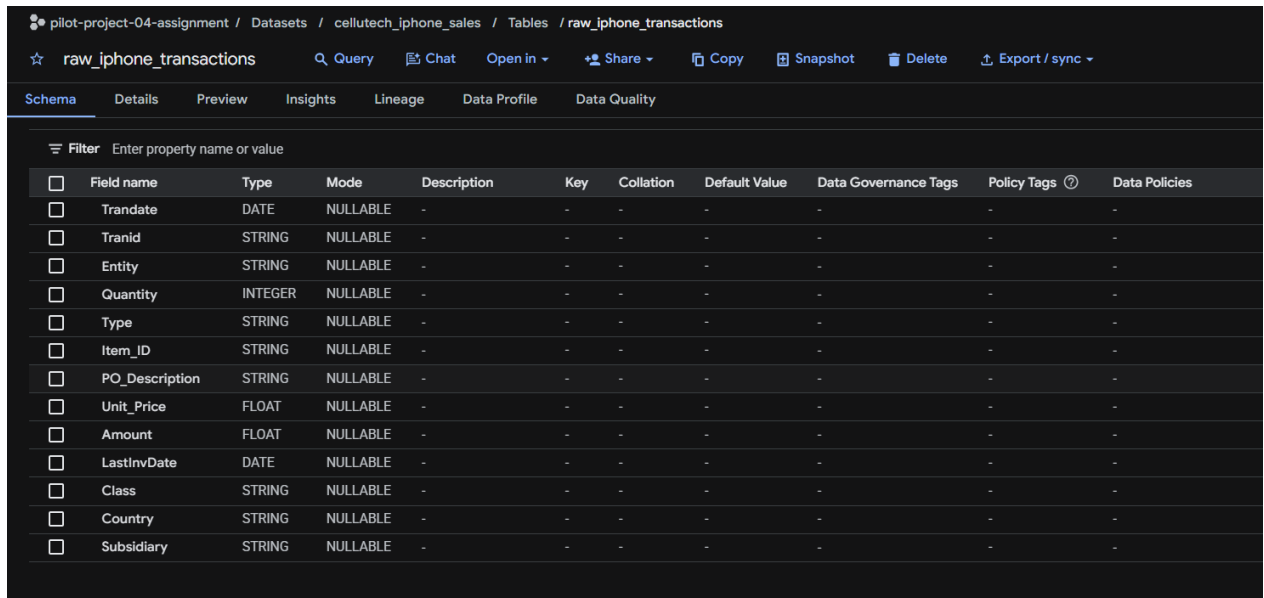
[SCREENSHOT: BigQuery Console — Dataset Overview showing cellutech_iphone_sales with all 3 objects]



The screenshot shows the BigQuery Console interface for the dataset 'cellutech_iphone_sales'. The breadcrumb path is 'pilot-project-04-assignment / Datasets / cellutech_iphone_sales'. The dataset name 'cellutech_iphone_sales' is highlighted. There are buttons for 'Create Table', 'Chat', and 'Preview'. The 'Overview' tab is selected, showing a list of tables and routines. The table 'raw_iphone_transactions' is highlighted.

Table ID	Type	Create time	Expiration time	Label	
raw_iphone_transactions	Table	Sep 24, 2026, 11:54:51 AM UTC+5	Nov 23, 2026, 11:54:51 AM UTC+5	None	: ☆
raw_product_master	Table	Sep 25, 2026, 9:26:29 PM UTC+5	Nov 24, 2026, 9:26:29 PM UTC+5	None	: ☆
rpt_iphone_sales	View	Sep 28, 2026, 10:32:41 PM UTC+5	Nov 27, 2026, 10:32:41 PM UTC+5	None	: ☆

[SCREENSHOT: BigQuery Console — raw_iphone_transactions Schema tab with column names and data types]



The screenshot shows the BigQuery Console interface for the table 'raw_iphone_transactions'. The breadcrumb path is 'pilot-project-04-assignment / Datasets / cellutech_iphone_sales / Tables / raw_iphone_transactions'. The table name 'raw_iphone_transactions' is highlighted. There are buttons for 'Query', 'Chat', 'Open in', 'Share', 'Copy', 'Snapshot', 'Delete', and 'Export / sync'. The 'Schema' tab is selected, showing a list of columns with their names, types, modes, descriptions, keys, collations, default values, data governance tags, policy tags, and data policies.

Field name	Type	Mode	Description	Key	Collation	Default Value	Data Governance Tags	Policy Tags	Data Policies
Trandate	DATE	NULLABLE	-	-	-	-	-	-	-
Tranid	STRING	NULLABLE	-	-	-	-	-	-	-
Entity	STRING	NULLABLE	-	-	-	-	-	-	-
Quantity	INTEGER	NULLABLE	-	-	-	-	-	-	-
Type	STRING	NULLABLE	-	-	-	-	-	-	-
Item_ID	STRING	NULLABLE	-	-	-	-	-	-	-
PO_Description	STRING	NULLABLE	-	-	-	-	-	-	-
Unit_Price	FLOAT	NULLABLE	-	-	-	-	-	-	-
Amount	FLOAT	NULLABLE	-	-	-	-	-	-	-
LastInvDate	DATE	NULLABLE	-	-	-	-	-	-	-
Class	STRING	NULLABLE	-	-	-	-	-	-	-
Country	STRING	NULLABLE	-	-	-	-	-	-	-
Subsidiary	STRING	NULLABLE	-	-	-	-	-	-	-

[SCREENSHOT: BigQuery Console — rpt_iphone_sales View Details showing 25 output columns]

pilot-project-04-assignment / Datasets / cellutech_iphone_sales / Tables / rpt_iphone_sales

☆ rpt_iphone_sales Query Chat Open in ▾ Share ▾ Copy Delete

Schema Details Insights Lineage Data Profile Data Quality

Filter Enter property name or value

☐	Field name	Type	Mode	Description	Key	Collation	Default Value	Data Governance Tags	Policy Tags ⓘ	Data Policies
☐	transaction_date	DATE	NULLABLE	-	-	-	-	-	-	-
☐	transaction_id	STRING	NULLABLE	-	-	-	-	-	-	-
☐	entity_name	STRING	NULLABLE	-	-	-	-	-	-	-
☐	quantity	INTEGER	NULLABLE	-	-	-	-	-	-	-
☐	transaction_type	STRING	NULLABLE	-	-	-	-	-	-	-
☐	item_id	STRING	NULLABLE	-	-	-	-	-	-	-
☐	product_description	STRING	NULLABLE	-	-	-	-	-	-	-
☐	unit_price	NUMERIC	NULLABLE	-	-	-	-	-	-	-
☐	amount	NUMERIC	NULLABLE	-	-	-	-	-	-	-
☐	last_invoice_date	DATE	NULLABLE	-	-	-	-	-	-	-
☐	sales_class	STRING	NULLABLE	-	-	-	-	-	-	-
☐	country	STRING	NULLABLE	-	-	-	-	-	-	-
☐	subsidiary	STRING	NULLABLE	-	-	-	-	-	-	-
☐	iphone_model	STRING	NULLABLE	-	-	-	-	-	-	-
☐	storage_capacity	STRING	NULLABLE	-	-	-	-	-	-	-
☐	color	STRING	NULLABLE	-	-	-	-	-	-	-
☐	transaction_year	INTEGER	NULLABLE	-	-	-	-	-	-	-
☐	transaction_month	INTEGER	NULLABLE	-	-	-	-	-	-	-
☐	transaction_quarter	INTEGER	NULLABLE	-	-	-	-	-	-	-
☐	year_month	STRING	NULLABLE	-	-	-	-	-	-	-
☐	month_name_year	STRING	NULLABLE	-	-	-	-	-	-	-
☐	day_of_week	STRING	NULLABLE	-	-	-	-	-	-	-
☐	days_since_last_invoice	INTEGER	NULLABLE	-	-	-	-	-	-	-
☐	transaction_category	STRING	NULLABLE	-	-	-	-	-	-	-
☐	price_tier	STRING	NULLABLE	-	-	-	-	-	-	-

[SCREENSHOT: BigQuery Console — raw_product_master Query Results showing sample rows]

pilot-project-04-assignment / Datasets / cellutech_iphone_sales / Tables / raw_product_master

☆ raw_product_master Query Chat Open in ▾ Share ▾ Copy Snapshot Delete Export / sync ▾

Schema Details Preview Insights Lineage Data Profile Data Quality

Filter Enter property name or value

☐	Field name	Type	Mode	Description	Key	Collation	Default Value	Data Governance Tags	Policy Tags ⓘ	Data Policies
☐	Item_ID	STRING	NULLABLE	-	-	-	-	-	-	-
☐	PO_Description	STRING	NULLABLE	-	-	-	-	-	-	-
☐	Model	STRING	NULLABLE	-	-	-	-	-	-	-
☐	Storage	STRING	NULLABLE	-	-	-	-	-	-	-
☐	Color	STRING	NULLABLE	-	-	-	-	-	-	-

Deliverable 2: SQL Scripts

All SQL scripts used for cleaning, formatting, transformations, and analysis given in:

File	Format	Contents
pilot_project_sql_scripts.sql	.sql	Raw SQL — can be directly copied and re-executed in BigQuery

2.1 SQL Techniques Demonstrated

SQL Technique	Usage in Project
CTEs (Common Table Expressions)	Two-stage CTE pipeline: base_transactions (cleaning) → enriched (JOINS + derived fields)
CASE Statements	Transaction categorization (Revenue vs Procurement), Price tier classification (4 tiers)
Date Functions	EXTRACT (year, month, quarter), FORMAT_DATE (year_month, month_name, day_of_week), DATE_DIFF
Calculated Fields	amount (Qty × Price recalculation), days_since_last_invoice
JOINS	LEFT JOIN with product_master to add Model, Storage, Color product dimensions
Type Casting	CAST to DATE, INT64, NUMERIC for proper analytical data types
String Functions	TRIM() applied on all string fields to remove whitespace artifacts
Window Functions	SUM() OVER() for percentage-of-total calculations in analytical queries
WHERE Filtering	NULL row exclusion for data sanitization (920 ghost rows removed)
Aggregations	GROUP BY with SUM, COUNT, AVG, COUNTIF, COUNT(DISTINCT) across 12 analytical queries
UNION ALL	Used in reconciliation queries to compare raw vs. reporting layer side-by-side

2.2 Script Structure

Section	Queries	Purpose
Section 1	7 queries	Schema review & data quality checks (row counts, null audit, distinct values, date range, join coverage)
Section 2	1 view	Main transformation — CREATE OR REPLACE VIEW with CTEs, CASE, JOINS, 25 output columns
Section 3	12 queries	Analytical queries (monthly trend, quarterly, entities, models, storage, country, subsidiary, types, reps, price tiers, colors, YoY)
Section 4	5 queries	Data validation & reconciliation (row counts, financial match, distinct values, duplicates, null check)
Section 5	2 queries	Executive KPI summary + Revenue vs Procurement split

Deliverable 3: Looker Studio Dashboard

An interactive executive management dashboard was built in Looker Studio, connected directly to the rpt_iphone_sales BigQuery reporting view.

3.1 Dashboard Title

Cellutech FZCO - Global iPhone Sales Dashboard

3.2 Dashboard Link

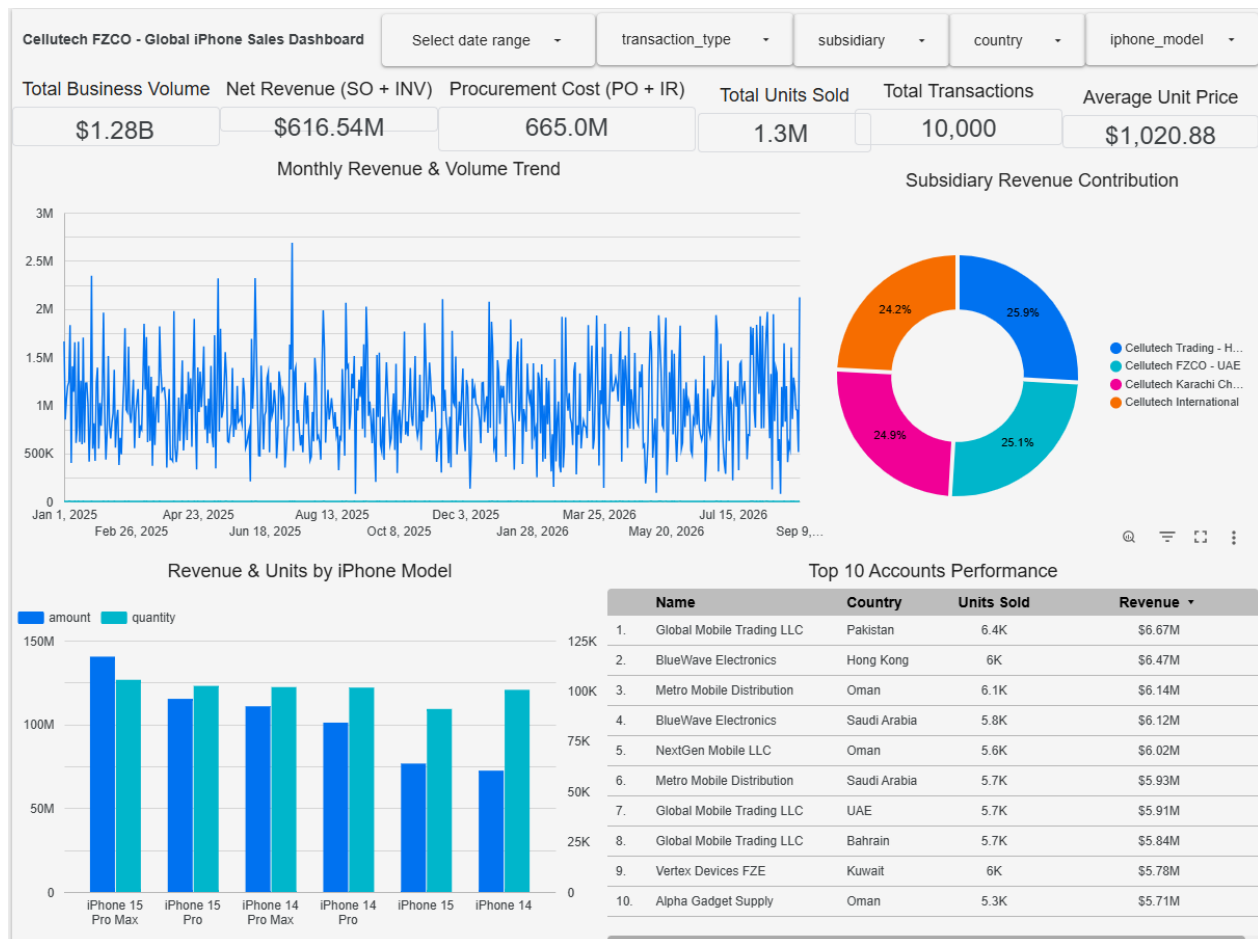
[<https://datastudio.google.com/reporting/174be905-0103-405b-be1f-d405acb45fdb>]

3.3 Dashboard Components

Component	Type	Description
Total Business Volume	Scorecard (KPI)	SUM of amount across all 4 transaction types — \$1.28B
Net Revenue (SO + INV)	Scorecard (KPI)	Filtered to Sales Order + Invoice only — \$616.54M (true revenue)
Procurement Cost (PO + IR)	Scorecard (KPI)	Filtered to Purchase Order + Item Receipt — \$665.0M (procurement cost)
Total Units Sold	Scorecard (KPI)	SUM of quantity — 1.3M units
Total Transactions	Scorecard (KPI)	COUNT of transaction records — 10,000
Average Unit Price	Scorecard (KPI)	AVG of unit_price — \$1,020.88
Monthly Revenue & Volume Trend	Time Series Chart	Monthly amount trend line — Jan 2025 to Sep 2026
Subsidiary Revenue Contribution	Donut Chart	Percentage contribution by each subsidiary (HK 25.9%, UAE 25.1%, Karachi 24.9%, International 24.2%)
Revenue & Units by iPhone Model	Bar Chart (Dual Axis)	Amount (left axis) and quantity (right axis) comparison by iPhone model series
Top 10 Accounts Performance	Table (Sorted)	Entity name, Country, Units Sold, Revenue — sorted by amount descending
Select date range	Interactive Filter	Date range selector at the top
transaction_type	Interactive Filter	Dropdown for 4 transaction types
subsidiary	Interactive Filter	Dropdown for 4 subsidiaries
country	Interactive Filter	Dropdown for 12 countries
iphone_model	Interactive Filter	Dropdown for 6 iPhone model series

3.4 Dashboard Screenshot

[SCREENSHOT: Full Looker Studio Dashboard — showing all KPI cards, charts, filters, and table]



Deliverable 4: Data Validation & Reconciliation

This section demonstrates that the SQL transformation preserved data integrity. All metrics match exactly between the raw staging layer and the reporting view, with zero financial variance.

4.1 Data Cleaning Summary

Issue Identified	Root Cause	Resolution
920 trailing blank rows in BigQuery raw table	The source Excel file contained cell formatting that extended beyond the 10,000 actual data rows. When uploaded to BigQuery, these formatted-but-empty cells were ingested as 920 completely null records.	Applied WHERE Tranid IS NOT NULL AND Trandate IS NOT NULL in the base CTE. Raw staging table preserved unchanged for audit traceability.
Amount column stored as Excel formulas	Excel source had =D2*H2 formulas instead of calculated values. During CSV conversion these were pre-computed.	Explicitly recalculated in SQL as ROUND(Quantity * Unit_Price, 2) for transparency and independent verification.
String whitespace artifacts	Potential leading/trailing whitespace in text fields from Excel export.	Applied TRIM() to all string columns (Entity, Type, Item_ID, Class, Country, Subsidiary) in the base CTE.

4.2 Reconciliation Summary

Metric	Raw Staging Table	Reporting View	Variance	Status
Raw Line Count	10,920	10,000	-920 (blank rows filtered)	PASS
Valid Transactions	10,000	10,000	0	PASS
Total Quantity	1,255,021	1,255,021	0	PASS
Total Amount (\$)	\$1,281,559,743.78	\$1,281,559,743.78	\$0.00	PASS
Distinct SKUs	93	93	0	PASS
Distinct Entities	12	12	0	PASS
Distinct Countries	12	12	0	PASS
Distinct Subsidiaries	4	4	0	PASS
Duplicate Transaction IDs	0	0	0	PASS

4.3 Reconciliation Query Screenshots

[SCREENSHOT: BigQuery — Query 4.1: Row count (Raw 10,920 total / 10,000 valid / 920 blanks vs View 10,000)]

Query results					
Job information		Results	Visualization	JSON	Execution details
		Execution graph			
Row	layer	total_records	valid_records	blank_ghost_rows	
1	Raw Table (incl. Excel Trailing B...	10920	10000	920	
2	Reporting View (Sanitized)	10000	10000	0	

[SCREENSHOT: BigQuery — Query 4.2: Financial reconciliation (identical quantity and amount totals)]

Query results				
Job information		Results	Visualization	JSON
		Execution details	Execution graph	
Row	source	total_quantity	total_amount	
1	Raw Table (Valid Rows Only)	1255021	1281559743.78	
2	Reporting View (Cleaned)	1255021	1281559743.78	

Deliverable 5: Business Findings & Strategic Recommendations

The following 6 findings are derived directly from the SQL analysis queries and are supported by the Looker Studio dashboard. The assignment requires at least 3 findings; we identified 6 to demonstrate deeper analytical thinking.

Finding 1: Revenue vs. Procurement Imbalance — Working Capital Risk

Observation:

The dataset contains 4 transaction types that fall into two business categories. By classifying them using a CASE statement (Sales Order + Invoice = Revenue; Purchase Order + Item Receipt = Procurement), we identified that procurement volume exceeds revenue volume by \$48.5M, a 7.8% gap.

Category	Transaction Types	Amount	Share
Revenue	Sales Order + Invoice	\$616,540,488.90	48.1%
Procurement	Purchase Order + Item Receipt	\$665,019,254.88	51.9%
Total Business Volume	All 4 types combined	\$1,281,559,743.78	100%

Strategic Recommendation:

The procurement-over-revenue gap suggests either inventory buildup, pricing compression on the sales side, or advance purchasing ahead of new product launches. Management should monitor this gap monthly to prevent working capital strain and ensure sell-through rates keep pace with purchasing velocity.

Supporting SQL: Query 3.8 (Transaction Type Breakdown) + Query 5.2 (Revenue vs Procurement Split)

[SCREENSHOT: Dashboard KPI cards showing Total Business Volume, Net Revenue, and Procurement Cost]

Cellutech FZCO - Global iPhone Sales Dashboard		Select date range ▾	transaction_type ▾	subsidi
Total Business Volume	Net Revenue (SO + INV)	Procurement Cost (PO + IR)	Total Units Sold	
\$1.28B	\$616.54M	665.0M	1.3M	

Finding 2: Premium iPhone Models (Pro/Pro Max) Command Disproportionate Revenue Share

Observation:

iPhone 15 Pro Max and iPhone 15 Pro lead all 6 model series in total dollar volume, followed closely by iPhone 14 Pro Max and iPhone 14 Pro. The Pro/Pro Max models collectively drive approximately 60-65% of total business volume despite representing a smaller share of total unit sales. This is driven by their higher Average Selling Price (ASP > \$1,200 for Pro Max vs. \$575-\$800 for standard models).

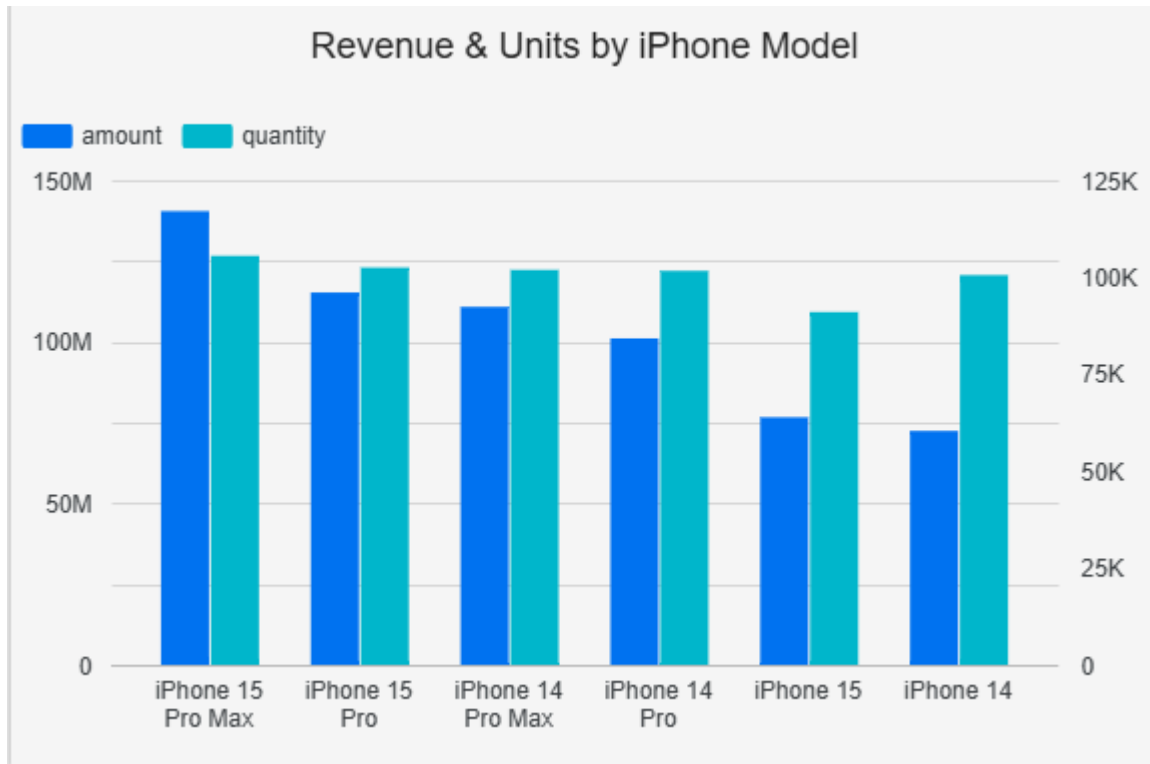
Data Points from Dashboard:

iPhone Model	Volume Rank	Observation
iPhone 15 Pro Max	#1	Highest amount — driven by premium ASP and strong demand
iPhone 15 Pro	#2	Close second — Pro tier remains the high-value workhorse
iPhone 14 Pro Max	#3	Previous generation Pro Max still commands strong volume
iPhone 14 Pro	#4	Consistent demand for Pro tier across both generations
iPhone 15 (Standard)	#5	Lower ASP limits dollar contribution despite healthy unit sales
iPhone 14 (Standard)	#6	Lowest volume — aging base model

Strategic Recommendation:

Maintain aggressive safety stock on Pro Max 256GB/512GB variants (highest revenue SKUs). Consider trade-up incentive programs for entities purchasing standard models to shift mix toward higher-margin Pro/Pro Max tiers. Monitor iPhone 14 standard model demand decline and optimize procurement to avoid overstocking.

[SCREENSHOT: Dashboard — 'Revenue & Units by iPhone Model' bar chart]



Finding 3: Sales Rep (Class) Workload is Heavily Skewed — Rebalancing Needed

Observation:

19 sales representatives handle the 10,000 transactions, but workload distribution is heavily skewed. The top representative (George) handles 1,057 transactions (10.6%), while the lowest (Mazhar) handles only 37 (0.4%). The top 6 reps collectively manage 51.9% of all transactions, while the bottom 7 reps collectively manage only 15.5%.

Sales Rep (Class)	Transactions	Workload %	Tier
George	1,057	10.6%	Heavy
John	841	8.4%	Heavy
Saqib	841	8.4%	Heavy
Shayan	841	8.4%	Heavy
Ismail	824	8.2%	Heavy
Ahsan	787	7.9%	Heavy
Julee / Adil / Ali / Sara / May	~494 each	~4.9% each	Moderate
David / Natalia / Tokio / Khasif	~350 each	~3.5% each	Light
Angle / Lee	~313 each	~3.1% each	Light
Jhanazab	276	2.8%	Light
Mazhar	37	0.4%	Minimal

Strategic Recommendation:

Management should review territory and account assignments to redistribute workload. Mazhar at 37 transactions (0.4%) is dramatically underutilized compared to George at 1,057 (10.6%). Rebalancing could improve response times, reduce burnout risk for top performers, and improve overall client satisfaction and account coverage.

Supporting SQL: Query 3.9 (Sales Class Performance & Workload Distribution) — includes pct_workload column

Finding 4: Subsidiary Contribution is Remarkably Balanced — Strong Geographic Diversification

Observation:

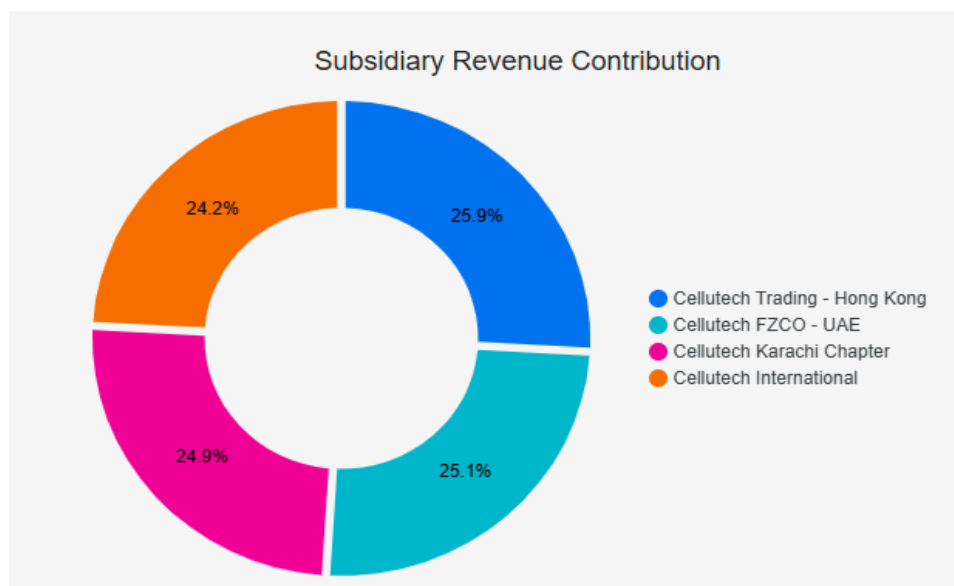
All 4 Cellutech subsidiaries contribute nearly equally to total business volume, with no single subsidiary exceeding 26% or falling below 24%. This even distribution across geographic fulfillment channels represents strong diversification, reducing single-region dependency risk.

Subsidiary	Contribution %	Assessment
Cellutech Trading - Hong Kong	25.9%	Leads slightly — Asia-Pacific hub
Cellutech FZCO - UAE	25.1%	Core Middle East hub
Cellutech Karachi Chapter	24.9%	South Asia operations
Cellutech International	24.2%	Global / multi-region

Strategic Recommendation:

The balanced subsidiary contribution is a positive indicator of geographic diversification. However, management should monitor whether this balance is organic or if it reflects artificial allocation policies. If organic, this represents a resilient distribution network. If artificial, there may be opportunity to optimize by routing more volume through the most cost-efficient subsidiary channels.

[SCREENSHOT: Dashboard — 'Subsidiary Revenue Contribution' donut chart showing 25.9% / 25.1% / 24.9% / 24.2%]



Appendix A: Data Cleaning Notes & Assumptions

As required by the assignment rules: 'Do not change business values simply to make the dashboard look correct.' All cleaning consisted of null row exclusion, type casting, whitespace trimming, and field derivation. Original values are preserved in the raw staging table for audit traceability.

Trailing Blank Rows (920 Null Records)

The source Excel file contained cell formatting that extended beyond the 10,000 actual data rows. When loaded into BigQuery, these trailing formatted-but-empty cells were ingested as 920 completely null records (rows 10,001–10,920). All 13 columns are NULL in these rows, confirming they are formatting artifacts, not partial records. These were filtered using WHERE Tranid IS NOT NULL AND Trandate IS NOT NULL in the reporting view CTE. The raw staging table was intentionally left unchanged for full audit traceability.

Amount Column (Excel Formulas)

The Amount column in the source file contained Excel formulas (=D2*H2, =D3*H3, etc.) rather than static calculated values. During the CSV conversion step, these formulas were evaluated. In the SQL reporting view, the amount is explicitly recalculated as ROUND(Quantity * Unit_Price, 2) for transparency and independent verification.

Transaction Type Classification

The 4 transaction types were classified into 2 business categories using a CASE statement: Revenue (Sales Order + Invoice) represents customer-facing sales activity; Procurement (Purchase Order + Item Receipt) represents supplier-facing purchasing activity. This enables separate analysis of sell-through vs. buy-in metrics and prevents misleading the total business volume as 'revenue.'

Entity-Country Many-to-Many Relationship

Data profiling revealed that all 12 entities transact across all 12 countries and all 4 subsidiaries. This indicates that the Country field represents the destination market rather than entity domicile, and Subsidiary represents the fulfillment channel. Both are independent analytical dimensions, not entity attributes.

LEFT JOIN Strategy

A LEFT JOIN was used between transactions and the product master to prevent silent data loss in case of any unmatched Item_IDs. In this dataset, all 93 SKUs matched perfectly (0 orphans), making the LEFT JOIN result identical to an INNER JOIN — but LEFT JOIN is the safer analytical practice for production pipelines.

Explicit Column Listing (No SELECT *)

Per the assignment requirement to 'avoid unnecessary SELECT * in the final reporting query,' the final view explicitly lists all 25 output columns instead of using SELECT *. This provides clear documentation of the output schema and prevents unintended column exposure.

No Business Values Changed

Per the assignment rules, no business values were altered. All cleaning consisted solely of null row exclusion, type casting, whitespace trimming (TRIM), and derived field creation. Original values are preserved unchanged in the raw staging table.

— *End of Submission Document* —